

ITERATION 2:

Objective

Convert your **static UI prototype (Iteration 1)** into a **fully working game for Level 1**.

At this stage, students must implement:

- Real-time gameplay
 - Physics (ball movement + collision)
 - Input handling
 - Score + lives system
 - Full integration with menu and screens
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SCOPE OF ITERATION 2

You must ensure:

- “Start Game” from menu launches **Level 1**
- Game is **playable from start to finish**
- Player can:
 - Control paddle
 - Break bricks
 - Lose lives
 - Reach game over

Only **Level 1** is required here. No multi-level system yet.

DETAILED MARKING BREAKDOWN (40 MARKS)

1. INPUT HANDLING (5 Marks)

Requirements

- Paddle controlled via keyboard (← → or A/D)
- Menu navigation must still work
- Name input from Iteration 1 must remain functional

Marking

Component	Marks
Paddle movement input	2
Menu navigation working	1
Name input retained	1
No input lag/glitches	1

2. BALL MECHANICS (10 Marks)

Requirements

- Ball must move continuously
- Direction updates dynamically
- Speed controlled via delay/timer

Marking

Component	Marks
Continuous movement	2
Wall collision (correct reflection)	2
Paddle collision (angle/direction change)	3
Brick collision detection	3

If ball passes through objects → marks deducted here directly

3. PADDLE MECHANICS (5 Marks)

Requirements

- Smooth left/right movement
- Cannot move beyond screen boundaries

Marking

Component	Marks
Smooth movement	2
Boundary restriction	2
Reasonable speed control	1

4. BRICK SYSTEM (6 Marks)

Requirements

- Brick disappears when hit
- Each brick increases score
- No duplicate collision issues

Marking

Component	Marks
Brick removal	2
Score increment	2
Accurate collision (no ghost hits)	2

5. GAME HUD (DISPLAY SYSTEM) (4 Marks)

Requirements

HUD must update **in real time**:

- Score
- Lives
- Username
- Level (should show Level 1)

Marking

Component	Marks
Score updates correctly	1
Lives display updates	1
Username visible	1
Level indicator working	1

6. LIVES SYSTEM (5 Marks)

Requirements

- Player starts with **3 lives**
- Ball missing paddle reduces life
- Ball resets after life loss
- Game ends at 0 lives

Marking

Component	Marks
Initial lives = 3	1
Life decrement logic	1
Ball reset after miss	1
Game over trigger	2

7. GAME FLOW INTEGRATION (5 Marks)

Requirements

- Menu → Game → Game Over → Menu
- No crashes or freezes

Marking

Component	Marks
Start Game launches Level 1	2
Game Over screen appears	1
Restart / return to menu	1
Stability (no crash)	1

TECHNICAL EXPECTATIONS

You must demonstrate:

- Use of **procedures** (no monolithic code)
- Basic **game loop structure**
- Controlled timing using:
 - Delay loops OR
 - BIOS interrupt (INT 15h)

REQUIRED GAME LOOP

Every frame must logically follow:

1. Read keyboard input
2. Update paddle position
3. Update ball position

4. Check collisions:
 - Walls
 - Paddle
 - Bricks
 5. Update score and lives
 6. Redraw screen (ball, paddle, bricks, HUD)
 7. Apply delay (control speed)
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SUBMISSION REQUIREMENTS

Students must submit:

1. Code

- Fully working Level 1
- Proper modular structure

2. Documentation

Must include:

- Game loop explanation
- Collision handling logic
- Lives system explanation

3. Video Demo (3–6 minutes)

Must show:

- Menu navigation
- Starting game
- Paddle control
- Brick breaking
- Losing lives
- Game over